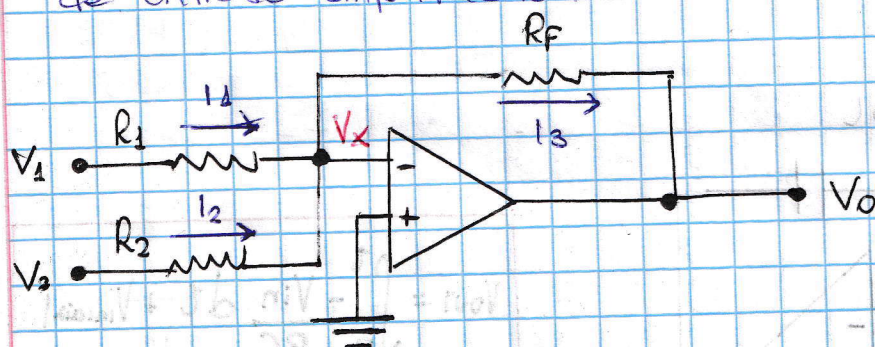


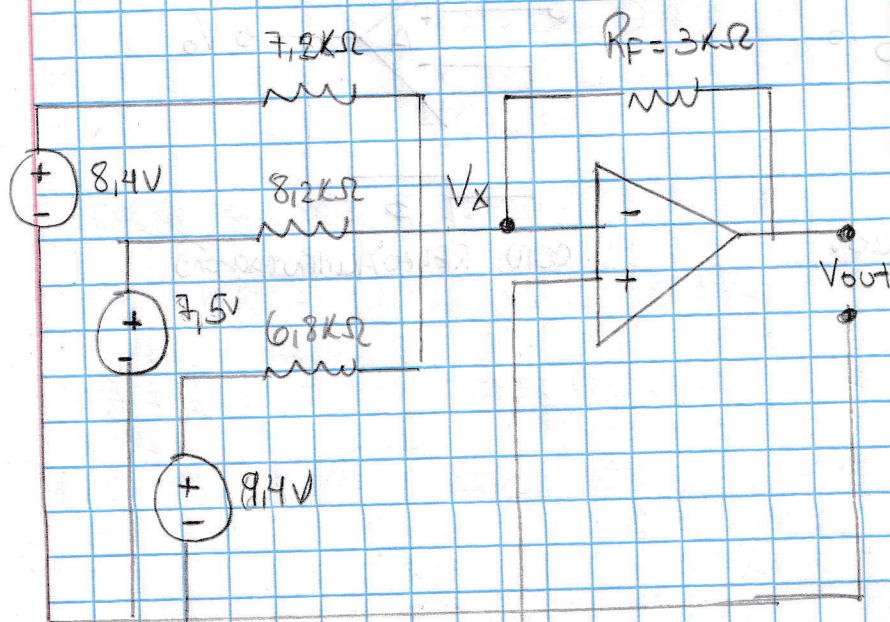
A) AMPLIFICADOR SUMADOR INVERSOR

Tiene dos o más voltajes conectadas a la entrada inversora del Amplificador Operacional (Op Amp) y la entrada no inversora esté conectada a tierra. Con esta configuración obtendremos el voltaje inverso de la suma de los voltajes de entrada amplificados.



$$V_0 = -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

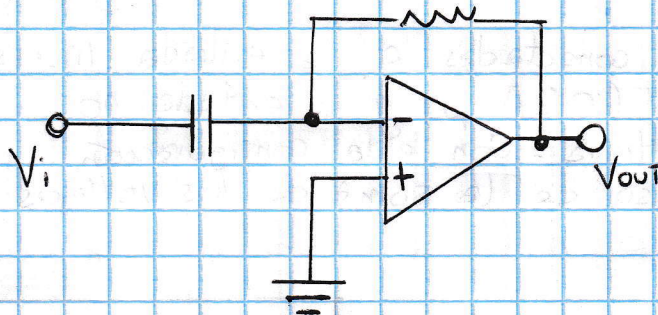
Ejm: $R_1 = 7,2 K\Omega$ $V_1 = 8,4V$ $R_F = 3 K\Omega$
 $R_2 = 8,2 K\Omega$ $V_2 = 7,5V$
 $R_3 = 6,8 K\Omega$ $V_3 = 9,4V$



$$V_0 = -3 \left[\frac{8,4}{7,2} + \frac{7,5}{8,2} + \frac{9,4}{6,8} \right]$$

$$V_0 = -10,38V$$

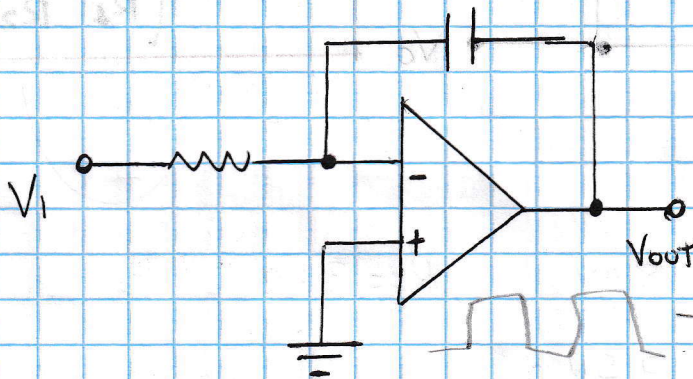
③ DERIVADOR IDEAL



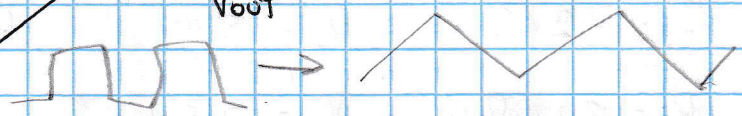
$$V_{out} = -RC \frac{d(V_{in})}{dt}$$



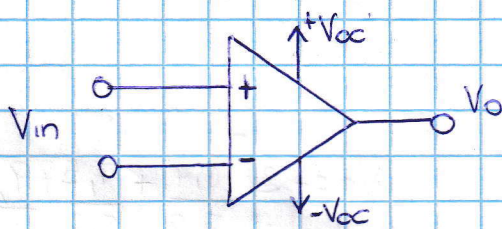
④ INTEGRADOR IDEAL



$$V_{out} = \int_0^t -\frac{V_{in}}{RC} dt + V_{inicial}$$

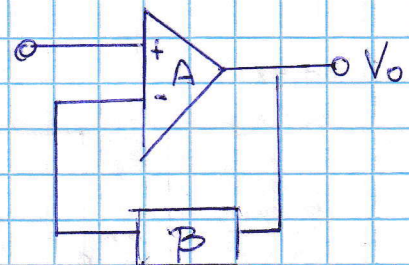


LAZO ABIERTO



SIN RETROALIMENTACIÓN

LAZO CERRADO



CON RETROALIMENTACIÓN

Sistema Numérico

CONVERSIÓN DE DECIMAL A BINARIO

	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1584	1	1	0	0	0	1	1	0	0	0	0

1024 ✓

560

512 ✓

48

32 ✓

16

16 ✓

-

$$1584 = 11000110000_2$$

SUMA BINARIA

ACARREOS : 1100

$$\begin{array}{r} 11110+ \\ 1100 \\ \hline 101010 \end{array}$$

RESTA BINARIA

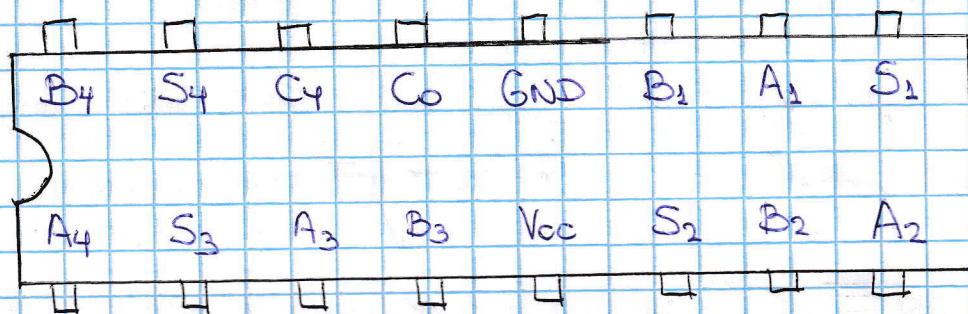
$$B = 1584 = 11000110000_2$$

$$A = 1841 = 11100110001_2$$

$$\textcircled{1} B - A = A + B_{C2} : \begin{array}{r} 11100110001 \dots A \\ 00110100000 \dots B_{C2} \\ \hline (-) 1100100000001 \end{array} \quad \begin{array}{l} (+) \\ (-) \end{array}$$

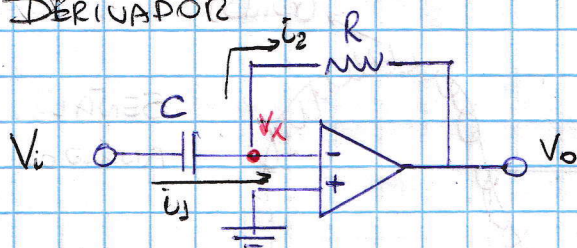
$$\begin{array}{r}
 \textcircled{2} \quad B - A = A_{c2} + B : \begin{array}{cccccccc} 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & \dots \end{array} A_{c2} \\
 \begin{array}{cccccccc} 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & \dots \end{array} B \\
 \hline
 (-) \quad \begin{array}{cccccccc} 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & \dots \end{array}
 \end{array}$$

DIAGRAMA SUMADOR 7483



$$\begin{array}{r}
 B = \begin{array}{cccc} B_4 & B_3 & B_2 & B_1 \end{array} \\
 A = \begin{array}{cccc} A_4 & A_3 & A_2 & A_1 \end{array} \\
 \hline
 \begin{array}{cccc} C_4 & S_4 & S_3 & S_2 & S_1 \end{array}
 \end{array}$$

• DERIVADOR



$$i_c = C \frac{d(V_i)}{dt}$$

$$i_1 = i_2$$

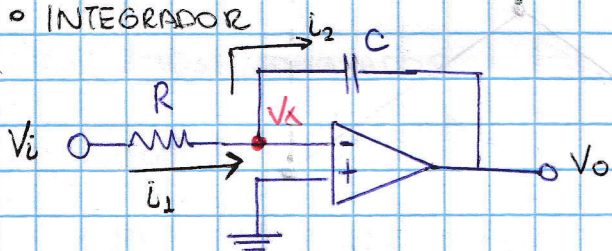
$$i_c = i_2$$

$$C \frac{d(V_i - V_x)}{dt} = \frac{V_x - V_o}{R}$$

$$C \frac{d(V_i)}{dt} = -\frac{V_o}{R}$$

$$V_o = -RC \frac{d(V_i)}{dt}$$

• INTEGRADOR



$$i_c = C \frac{d(V)}{dt}$$

$$i_1 = i_2$$

$$\frac{V_i - V_x}{R} = i_c$$

$$\frac{V_i - V_x}{R} = C \frac{d(V_x - V_o)}{dt}$$

$$\int \frac{V_i}{-RC} dt = \int \frac{d(V_o)}{dt} dt$$

$$V_o = -\frac{1}{RC} \int (V_i) dt$$

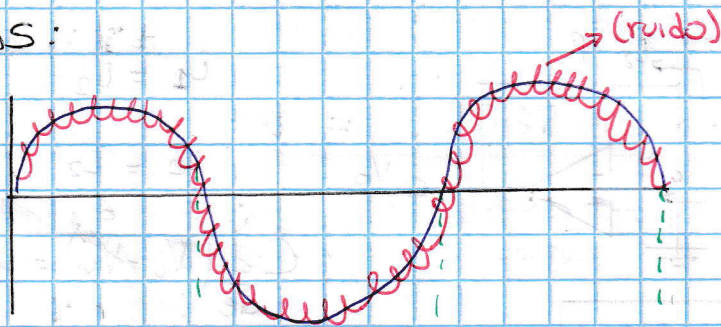
Ejercicio: $C = 10 \mu F$, $R = 1,2 k\Omega$, $V = 5,2 V$, $t = 15 ms$ (INTEGRADOR)

$$V_o = -\frac{1}{RC} \int V_i dt$$

$$V_o = \frac{-1}{(1,2 \cdot 10^3)(10 \cdot 10^{-6})} (5,2)(15 \cdot 10^{-3} - 0) = -6,5 V //$$

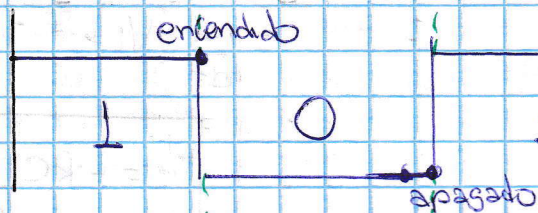
TIPOS DE ONDAS:

ONDA SENOIDAL



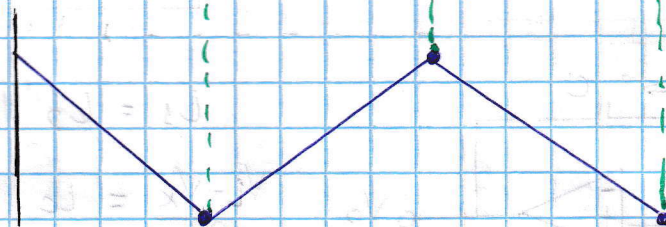
SEÑAL
ANÁLOGA

ONDA CUADRADA



SEÑAL
DIGITAL

ONDA TRIANGULAR



FUNCIONES LÓGICAS

El Álgebra Booleana es una herramienta diseñada para poder representar proposiciones lógicas en forma algebraica

Variable: Símbolo que representa magnitudes lógicas, (0 ó 1) (A)

Complemento: El inverso de la variable ($\bar{A}$)

SUMA (OR)

$$\begin{aligned} 0 + 0 &= 0 \\ 0 + 1 &= 1 \\ 1 + 0 &= 1 \\ 1 + 1 &= 1 \end{aligned}$$



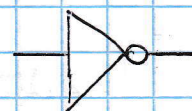
MULTIPLICACIÓN (AND)

$$\begin{aligned} 0 \cdot 0 &= 0 \\ 0 \cdot 1 &= 0 \\ 1 \cdot 0 &= 0 \\ 1 \cdot 1 &= 1 \end{aligned}$$



NEGACIÓN (NOT)

$$\begin{aligned} \bar{0} &= 1 \\ \bar{1} &= 0 \end{aligned}$$



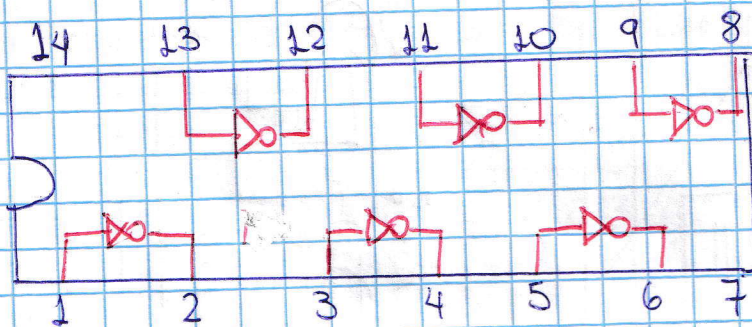
Ejercicio: $F(x, y, z) = x\bar{y} + y\bar{z} + z\bar{x}$

2^n combinaciones, $2^3 = 8$

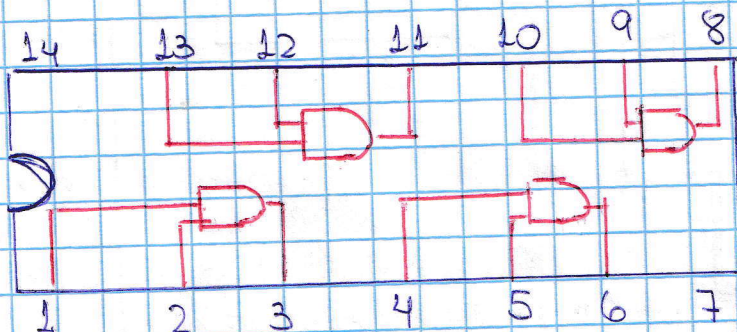
	X	Y	Z	R
0:	0	0	0	$0.1 + 0.1 + 0.1 = 0$
1:	0	0	1	$0.1 + 0.0 + 1.1 = 1$
2:	0	1	0	$0.0 + 1.1 + 0.1 = 1$
3:	0	1	1	$0.0 + 1.0 + 1.1 = 1$
4:	1	0	0	$1.1 + 0.1 + 0.0 = 1$
5:	1	0	1	$1.1 + 0.0 + 1.0 = 1$
6:	1	1	0	$1.0 + 1.1 + 0.0 = 1$
7:	1	1	1	$1.0 + 1.0 + 1.0 = 0$

0 CIRCUITOS INTEGRADOS

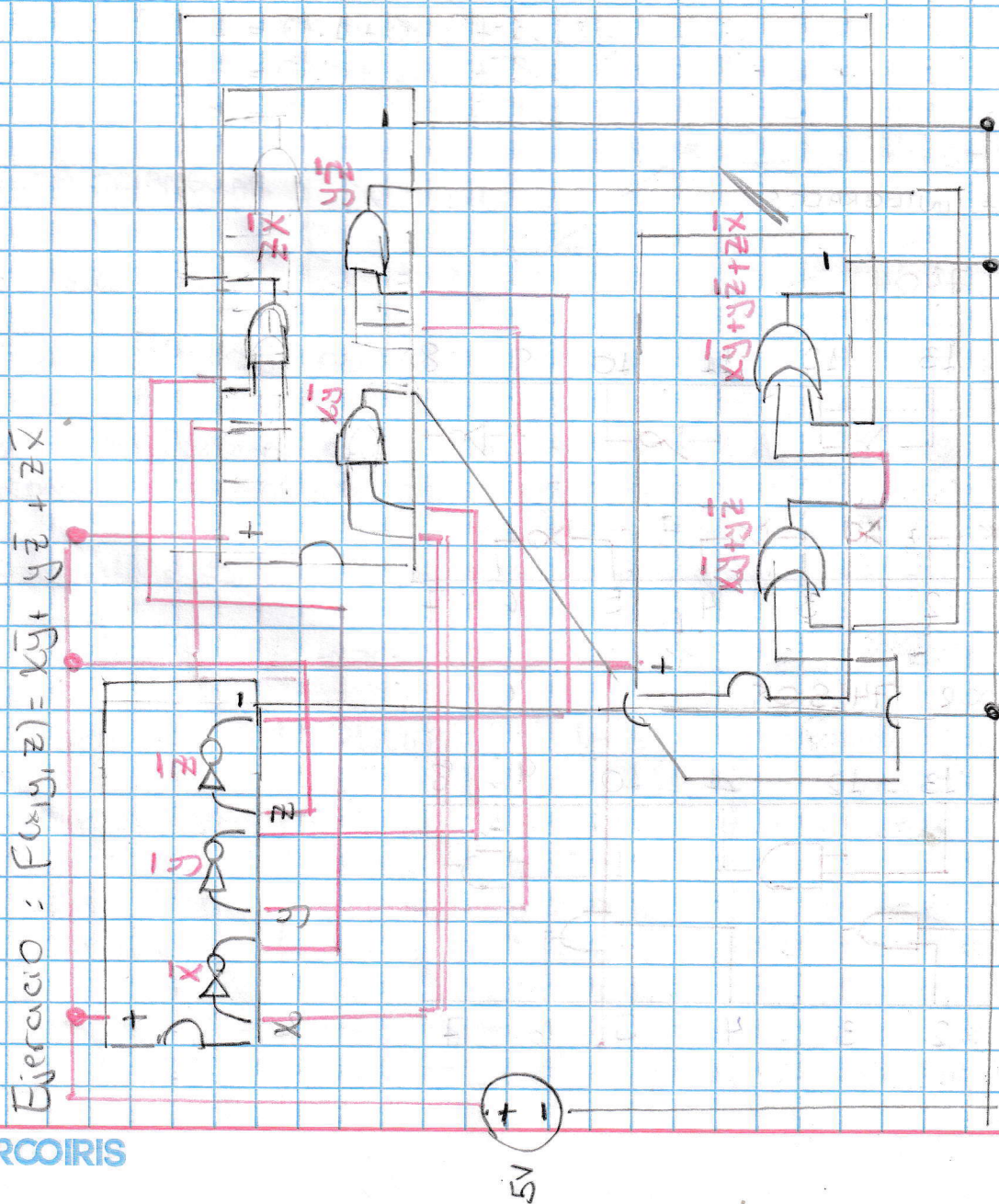
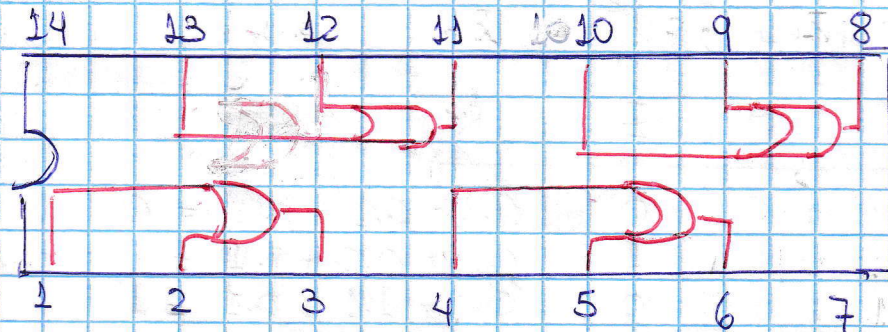
INVERSOR 7404



MULTIPLICADOR 74LS08



SOMADOR 74LS32



OPAMP: 7404

08

32

	X	Y	Z	$X\bar{Y}$	$Y\bar{Z}$	$Z\bar{X}$	$X\bar{Y} + Y\bar{Z}$	$F(x,y,z)$
PINES:	1	3	5	3	6	11	3	6
0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0	1
2	0	1	0	0	1	0	1	1
3	0	1	1	0	0	1	0	1
4	1	0	0	1	0	0	1	1
5	1	0	1	1	0	0	1	1
6	1	1	0	0	1	0	1	1
7	1	1	1	0	0	0	0	0

MAPA DE KARNAUGH

Herramienta utilizada para la simplificación de funciones lógicas booleanas y a diferencia de la resolución por álgebra de Boole, este es un método gráfico que implica conocer las representaciones canónicas de las funciones

$F = X\bar{Y} + Y\bar{Z} + Z\bar{X}$

Y \ XZ	X		$\bar{X}$		Y
	XZ	$\bar{X}\bar{Z}$	XZ	$\bar{X}\bar{Z}$	
1	6	7	3	2	1
	1	0	1	1	
2	4	5	1	0	1
	1	1	1	0	
	$\bar{Z}$	Z	Z	$\bar{Z}$	

Forma Canónica:

$$F(x,y,z) = \sum (1, 2, 3, 4, 5, 6) = \prod (0, 7)$$

SUMA DE PRODUCTOS DE TÉRMINOS
MÍNIMOS (SPT_{MIN})
MIN TÉRMINOS

PRODUCTO DE SUMA DE
TÉRMINOS MÁXIMOS (PST_{MAX})
MAX TÉRMINOS

Ejercicios:

① Se tiene $F(M, N, P, Q) = M\bar{N}Q + \bar{Q}P\bar{N} + \bar{P}\bar{M}P = M\bar{N}Q + \bar{Q}P\bar{N}$

MN \ PQ	$\bar{Q}$		Q	
	00	01	11	10
$\bar{N}$	0	0	0	1
N	0	0	0	0
$\bar{M}$	1	1	0	0
M	0	1	1	1
$\bar{P}$				
P				

$$F(M, N, P, Q) =$$

$$\sum(2, 10, 11)$$

	M	N	P	Q	F
0:	0	0	0	0	0
1:	0	0	0	1	0
2:	0	0	1	0	1
3:	0	0	1	1	0
4:	0	1	0	0	0
5:	0	1	0	1	0
6:	0	1	1	0	0
7:	0	1	1	1	0
8:	1	0	0	0	0
9:	1	0	0	1	0
10:	1	0	1	0	1
11:	1	0	1	1	1
12:	1	1	0	0	0
13:	1	1	0	1	0
14:	1	1	1	0	0
15:	1	1	1	1	0

② Realizar la siguiente suma en binario

A = 9 E D 7 4

B = 7 D C 6 5

1º DÍGITO ($C_0 = 0$)

4 : 0 1 0 0 +

5 : 0 1 0 1

0 1 0 0 1

C_4

4º DÍGITO ($C_0 = 1$)

E : 1 1 1 0 +

D : 1 1 0 0

1 1 0 1 1

C_4

2º DÍGITO ($C_0 = 0$)

7 : 0 1 1 1 +

6 : 0 1 1 0

0 1 1 0 1

C_4

5º DÍGITO ($C_0 = 1$)

9 : 1 0 0 1 +

7 : 0 1 1 1

1 0 0 0 1

C_4

3º DÍGITO ($C_0 = 0$)

D : 1 1 0 1 +

C : 1 1 0 0

1 1 0 0 1

C_4

A+B =

1 0001 1011 1001 1101 1001